



Fig. 7: Heater coil having 3 legs, thermistor mounting, and Teflon cables

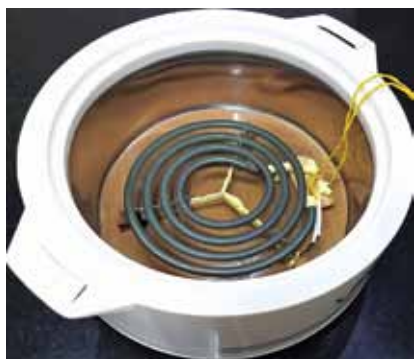


Fig. 8: Heater coil kept inside the casserole on a round piece of Teflon sheet

with a stainless steel surface inside (200mm dia × 80mm depth). Preferably keep a round piece of Teflon sheet below the heater coil.

Cover assembly. In Fig. 9, you will see the casserole cover assembly. In the centre of the cover, create a small hole and insert an 8mm diameter copper pipe. Secure the pipe permanently using Araldite to prevent steam from entering the insulated area inside the cover. This pipe allows the steam produced during cooking to escape, preventing water buildup inside the casserole.

Attach a 10mm thick Poly Urethane Foam (PUF) ring, as illustrated in Fig. 9(d), to the cover, as depicted in Fig. 9(a).

Place the soaked millets in a stainless-steel container. Add water, approximately 3 to 4 times the quantity of millets. Create a small hole in the container's cover, as depicted in Fig. 10. This hole allows the steam, generated during cooking, to escape through the copper pipe in the cover.

Usage. To cook, place the soaked millets and the required water in the stainless-steel box. Put the box inside the casserole on the heater coil, ensuring the copper pipe aligns with the hole. Connect the heater coils to H1 and H2 terminals, and the thermistor terminals to Th1 and Th2. Adjust the potentiometer knob for the desired cooking time.

Switch on S1; the white LED

lights up, but the heater remains off. To activate the heater, briefly press the S2 pushbutton. The green LED turns on, signifying the heater is running. At this point, the red LED remains off. After about 30 to 40 minutes, when the heater reaches the set temperature, the red LED begins to blink, indicating that the counter has started.

As time passes, steam gradually escapes through the copper pipe, and you can smell the cooking millets. When the counter overflows, the MOSFET switches off, and the green LED goes out. The buzzer sounds an alarm, signalling that the cooking is complete. If you need the cooked millets immediately, you can remove them, but if not, keep the cover closed. In this state, the food will stay warm for one to two hours. Before opening the cover, remember to turn off S1, and use gloves or a kitchen towel as the box will be hot.

Fig. 11 depicts the coil temperature variation over cooking time, resembling an elephant curve. During the initial 0 to 10 minutes, the coil temperature rises rapidly. Within this period, only the coil heats up.

After 10 minutes, heat begins transferring to the box, causing a slower temperature to increase up to

50 minutes due to the large quantity of water present. Beyond 50 minutes, as the water in the box gets absorbed by the millets, the temperature rises again. When the temperature surpasses 135°C, cooking is complete. This triggers the counter to overflow, and the MOSFET turns off. While the coil temperature decreases, the box temperature remains relatively high due to the insulation.

Energy consumption.

Power consumed by the coil = 78.4W

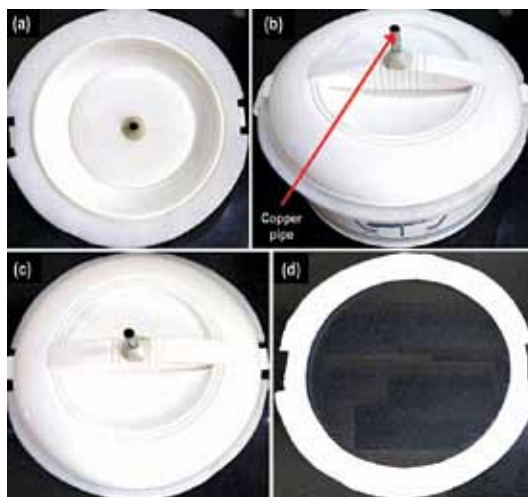


Fig. 9: (a) Cover with PUF ring, (b) full unit, (c) cover top view with copper pipe, and (d) PUF ring